

1. Introduction: must include,

- a. **Motivation:** Explanation of the reasons for choosing the project and its importance.
- b. **Problem Statement:** A clear and specific description of the problem the project aims to solve.
- c. **Goals of the Project:** A list of main and sub-goals, ensuring they are measurable.
- d. **Currently Available Solutions (Applications):** Review of similar existing applications or systems in the market.
- e. **Features Matrix:** A comparison table illustrating the features of current solutions versus the planned features of the proposed project.
- f. **Software Development Methodology Used:** Specify the methodology used (e.g., Waterfall, Agile, Scrum,...) with justification for its selection and how it will be applied in the project.

2. Requirements Analysis, document the process of investigating and defining system requirements

- a. **Requirements Gathering Tools:** Description of questionnaires, interviews, or other tools used.
- b. **Presentation of Results:** Present the results of the requirements gathering process using appropriate charts (bar charts, pie charts, etc.).
- c. **Discussion of Results:** Analysis and interpretation of the presented results, linking them to the project requirements.
- d. **Functional Requirements:** A detailed, numbered list of what the system must do.
- e. **Non-functional Requirements:** Such as performance, security, usability, compatibility.
- f. **Use Case Diagrams:** A visual representation of the interaction between users (actors) and the system.

3. Design, this section must present detailed system design:

- a. **Sequence Diagrams:** Illustrate the interaction of objects or components during the execution of a specific use case.
- b. **Entity-Relationship Diagram (ERD):** Describes the database structure and relationships between entities.

- c. Data Flow Diagrams (DFD): Level 0 (context diagram) and Level 1 (detailing main processes).
- d. State Diagram: (Optional) If the system contains entities with complex states.
- e. Algorithms or Pseudocode: (If available) For complex core processes.
- f. UML Class Diagrams: Shows the structure of objects in the system and their relationships.

4. Implementation Aspects, this section should clarify the practical aspects and next steps:

- a. Overall System Architecture: A block diagram illustrating the main components and their interconnections.
- b. Tools, Technologies, and Programming Languages: A list of the tools, programming languages, and frameworks to be used, indicated on the architecture diagram (Section 4.a).
- c. Prototype: Screenshots of the functional prototype with a brief description for each screen.

5. Future Work Plan (Plan for Senior Project 2)

(Suggested addition to link the report to the second part of the project)

- a. Remaining Tasks: A list of key tasks planned for completion in the Senior Project 2 semester.
- b. Proposed Timeline: A time estimate for completing the remaining tasks.
- c. Anticipated Risks and Challenges: Identification of potential challenges and mitigation plans.

References

- A list of sources and references consulted (books, research papers, websites, etc.).